

tests/dynamic/analyzers/ graceful_503_analyzer.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active self-validated analyzer (§11.4.107(10)). Signal 2 of 6 for the
dynamic data plane — graceful degradation.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

Signal 2: **branded 503 + Squid PID unchanged** (§11.4.69
network_connectivity; design §10/§13 graceful_503; §11.4.108
runtime-signature). When the target tunnel is DOWN, Squid must
return an intentional, **branded 503** (ERR_TUNNEL_DOWN body) **and
keep the same PID** across the request (it did not crash/restart).
It delegates the decision to the committed, self-tested
evidence.sh:assert_graceful_503 through that helper's documented
unit-test seams (EVIDENCE_503_CODE_OVERRIDE, EVIDENCE_503_BODY_FILE,
EVIDENCE_503_BODY_MARKER), feeding a captured probe manifest — so
the analyzer needs no live network.

Prerequisites

- Source library tests/dynamic/lib/analyzer_common.sh + the
committed tests/lib/evidence.sh.
- awk, POSIX sh.
- A probe manifest (key=val, one per line):
 - http_code=503
 - body_file=503_body.html (resolved relative to the manifest
dir)
 - pid_before=12345
 - pid_after=12345
 - marker=tunnel (optional branded-text marker)

Usage examples

Analyze a captured probe manifest:

```
tests/dynamic/analyzers/graceful_503_analyzer.sh analyze  
    <manifest-file>
```

Self-validate (golden-good PASS + golden-bad FAIL) – the default action:

```
tests/dynamic/analyzers/graceful_503_analyzer.sh --selftest
```

Edge cases

- **Manifest missing / evidence.sh not found** → FAIL, never a bluff.
- **PID changed across the request** (proxy crashed/restarted) → FAIL: a 503 from a crashed proxy is not graceful — the PID-unchanged check is the §11.4.108 runtime-signature that distinguishes them.
- **HTTP 200 instead of 503** (target leaked through) → FAIL.
- **Blank / unbranded 503 body** → FAIL (the branded marker is required).

§11.4.115 RED_MODE polarity

Consuming suites (chaos_suite.sh C1) run the §11.4.115 polarity: RED reproduces a crash/fail-open on the pre-fix stack; GREEN guards the branded-503 + PID-unchanged contract. This analyzer is the oracle each polarity cites.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean (`sh -n + bash -n`, §11.4.67).
- `_g503_manifest_get` parses `key=val` lines with `awk`; the body file is resolved relative to the manifest dir when not absolute.
- Drives `assert_graceful_503` through its env seams (no network) and forwards its rc (0 PASS / 1 FAIL).
- Self-test asserts: golden-good PASS; PID-changed, HTTP-200, blank-body → FAIL; missing-manifest negative → FAIL.

Related

- `tests/lib/evidence.sh` (`assert_graceful_503`) — the delegated oracle.
- `tests/dynamic/lib/analyzer_common.sh` — sourced base.
- `tests/dynamic/suites/chaos_suite.sh` — builds a manifest from the live capture and runs this analyzer.

- Fixtures: tests/dynamic/analyzers/fixtures/graceful_503/.
- Constitution §11.4.69 / §11.4.107 / §11.4.108 / §11.4.115; design §10/§13; research §4.

Last verified

2026-07-01 — self-test PASS (golden-good PASS + PID-changed/200/blank-body FAIL); sh -n + bash -n parse-clean. Live probe is exercised in **P10**.